STOCHASTIC AND DEEP LEARNING APPROACHES FOR EARTHQUAKE-INDUCED SOIL LIQUEFACTION ASSESSMENT

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I express my deepest gratitude to my supervisor **Dr. Arpit Jain** for discussions and guidance throughout the internship. Their suggestions allowed me to focus on the physical meaning of spatial variability, temporal dependence, earthquake input motion and pore-pressure response and move away from a view of stochastic modelling purely as random data generation.

I would also like to thank the researchers whose papers formed the backbone of this literature review and my family and friends for their continued support and encouragement throughout the internship.

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ABSTRACT

Earthquake-induced liquefaction is closely associated with excess pore-water pressure generation and the consequent reduction in effective stress in saturated sandy deposits. Dynamic centrifuge tests provide valuable measurements of this behaviour, but the data are available only at discrete sensor locations and are influenced by spatial variability, temporal dependence, measurement noise and variations in the applied shaking. The main objective of this summer internship was to review stochastic modelling methods used in similar geotechnical earthquake problems.

Seven papers were reviewed. Six are closely related to centrifuge testing, liquefiable sand, Gaussian-process methods, random fields, Monte Carlo simulation and seismic input variability. The seventh paper, by Raghu Kanth and Dash, gives a particularly clear example of converting sparse SPT observations into a large set of synthetic profiles through trend removal, an exponential Gaussian random field, spectral simulation and Monte Carlo liquefaction analysis. A flowchart was prepared for every paper so that the modelling sequence could be compared directly.

The internship also included a supporting study of artificial neural networks, convolutional neural networks and sequence models such as RNN, LSTM, BiLSTM and GRU, together with basic model-training practice on standard datasets. The combined review suggests that the present centrifuge dataset should be viewed as a spatiotemporal problem involving $u(x, y, z, t)$, $a(x, y, z, t)$ and the measured box excitation $a_{box}(t)$. A spatiotemporal Gaussian Process, potentially conditioned on the box motion and extended to joint pore-pressure and acceleration response, is a logical direction for the next stage. Gaussian random fields and Monte Carlo simulation provide complementary tools for spatial realizations and uncertainty propagation. No final synthetic-data model is claimed in this report; the internship establishes the literature-based methodology for that work.

Keywords: Liquefaction; Centrifuge test; Stochastic modelling; Gaussian Process; Gaussian Random Field; Monte Carlo Simulation; Pore-Water Pressure; Acceleration; Deep Learning

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ABBREVIATIONS

ANN: Artificial Neural Network

CNN: Convolutional Neural Network

RNN: Recurrent Neural Network

LSTM: Long Short-Term Memory

BiLSTM: Bidirectional Long Short-Term Memory

GRU: Gated Recurrent Unit

GP: Gaussian Process

GPR: Gaussian Process Regression

GRF: Gaussian Random Field

SGS: Sequential Gaussian Simulation

RFEM: Random Finite Element Method

MC: Monte Carlo

PPT: Pore-Pressure Transducer

EPWP: Excess Pore-Water Pressure

PGA: Peak Ground Acceleration

SPT: Standard Penetration Test

CPT: Cone Penetration Test

CSR: Cyclic Stress Ratio

FS: Factor of Safety

PWP: Pore-Water Pressure

PSD: Power Spectral Density

ACF: Autocorrelation Function

COV: Coefficient of Variation

FEM: Finite Element Method

ML: Machine Learning

DL: Deep Learning

CHAPTER 1

INTRODUCTION

1.1 Background and motivation

Soil liquefaction is a major earthquake-related geotechnical hazard, especially in saturated sandy deposits. Cyclic loading can rearrange the soil skeleton and build excess pore-water pressure. The effective-stress relation is $\sigma_v' = \sigma_v - u$. An increase in pore pressure can reduce effective stress and, consequently, shear resistance. The response can lead to settlement, lateral spreading, loss of bearing capacity and other forms of deformation.

Centrifuge testing gives a controlled way to observe this behaviour under dynamic loading. The present experiment uses a rectangular box filled with sand and contains four sensor levels. Each level contains six sensor positions involving pore-pressure transducers and accelerometers, while an additional accelerometer outside the soil box records the shaking of the box. The experiment is considered in two configurations: a horizontal arrangement and an inclined arrangement. The measured quantities are therefore distributed over space and time.

The central question was how uncertainty and correlation can be modelled so that synthetic observations and if generated later that can remain statistically consistent with the physical test. This led to a focused review of stochastic methods used in closely related geotechnical earthquake problems.

1.2 Objectives

- Review stochastic modelling studies related to liquefaction, dynamic centrifuge tests, spatial soil variability and earthquake input motion.
- Understand how Gaussian processes, random fields, Monte Carlo simulation and related methods are fitted to observations and used to produce predictions or stochastic realizations.
- Compare the reviewed methodologies with the present dataset containing $u(x, y, z, t)$, $a(x, y, z, t)$ and $a_{box}(t)$.
- Study ANN, CNN and sequence-based models as complementary data-driven tools for later stages of the work.
- Prepare a practical literature-derived framework for the next phase of the project.

1.3 Scope of the work

The work was mainly a literature and methodology study. The report explains the statistical logic used in the selected papers and maps it to the present centrifuge test. It does not claim that the final stochastic generator has already been calibrated on the actual sensor dataset. The machine learning part was used to build supporting knowledge about tabular, spatial and sequential modelling and to understand how a future synthetic dataset could be used responsibly.

CHAPTER 2

LITERATURE REVIEW

2.1 Liquefaction and seismic response

Liquefaction analysis involves both the cyclic loading and the response of the soil skeleton. Pore-pressure buildup is central to the phenomenon, while acceleration measurements describe the dynamic response through the deposit. This combination makes sensor-based centrifuge data especially suitable for stochastic analysis because there is a measurable input, a distributed response, and repeated observations in time.

2.2 Conventional, numerical and data-driven approaches

Conventional liquefaction assessment uses SPT, CPT or shear-wave velocity correlations and often reduces the problem to a factor of safety or probability of liquefaction. Effective-stress finite-element models provide richer information on excess pore pressure and deformation but can be computationally expensive. Machine learning offers a way to learn nonlinear response relationships from large collections of examples. In the internship, ANN was studied for tabular data, CNN for spatial patterns, and RNN, LSTM, BiLSTM and GRU for ordered data. These methods were considered as supporting tools rather than the main subject of the literature review.

2.3 Stochastic modelling background

The papers selected for detailed review can be grouped into three themes. GP studies learn probabilistic relationships between experimental variables and responses. Random-field studies represent soil heterogeneity explicitly and propagate it through numerical simulations. Other studies model variability in earthquake or base motion. The seventh paper shows how a deterministic trend and a stochastic residual field can be combined to create many synthetic soil profiles from sparse observations. This structure is particularly useful for understanding the present data-generation problem.

2.4 Paper 1 - Difference and Sensitivity Analyses of the LEAP-2017 Experiments

Goswami et al. (2020). Difference and Sensitivity Analyses of the LEAP-2017 Experiments.

The study uses LEAP-2017 centrifuge experiments on saturated, mildly sloping Ottawa F-65 sand. The authors compare acceleration histories from different realizations and break the

differences into interpretable quantities such as phase change, frequency shift and selected amplitude measures. The work is valuable here because it treats the variability among nominally similar centrifuge tests as information rather than as noise.

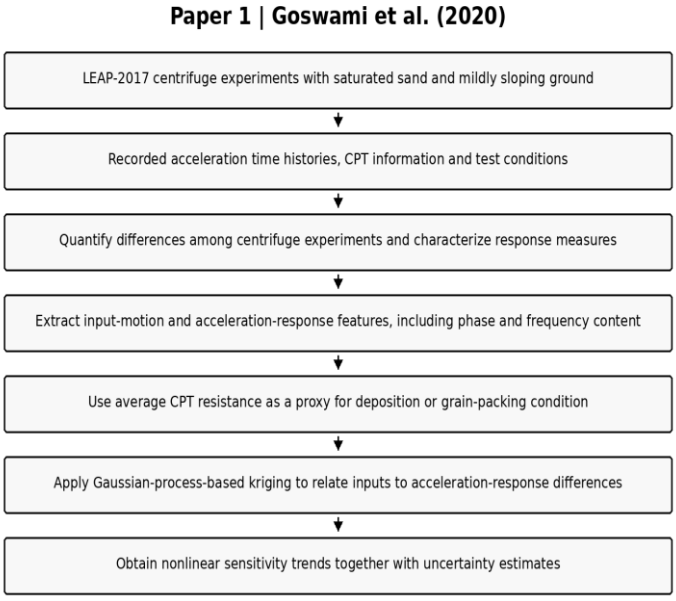


Figure 2.1 Methodology adopted in Paper 1 - Goswami et al. (2020)

Gaussian-process-based kriging is then used to examine nonlinear sensitivity relationships involving the input motion and the soil or deposition condition. The output is a response surface together with uncertainty. For the present experiment, the transferable idea is to extract useful features from $a_{box}(t)$ and relate them to acceleration or pore-pressure response at known sensor positions before attempting full time-history generation.

2.5 Paper 2 - Regression and trend analysis of LEAP 2020 experimental results

Zeghal et al. (2024). Regression and trend analysis of LEAP 2020 experimental results.

This paper analyses 23 LEAP-2020 centrifuge experiments on saturated Ottawa F-65 sand and relates test conditions and input-motion characteristics to measured response quantities. Pore-pressure buildup is one of the important response measures. The authors first check the consistency of the experimental dataset and then use Gaussian Process Regression to describe the response trends.

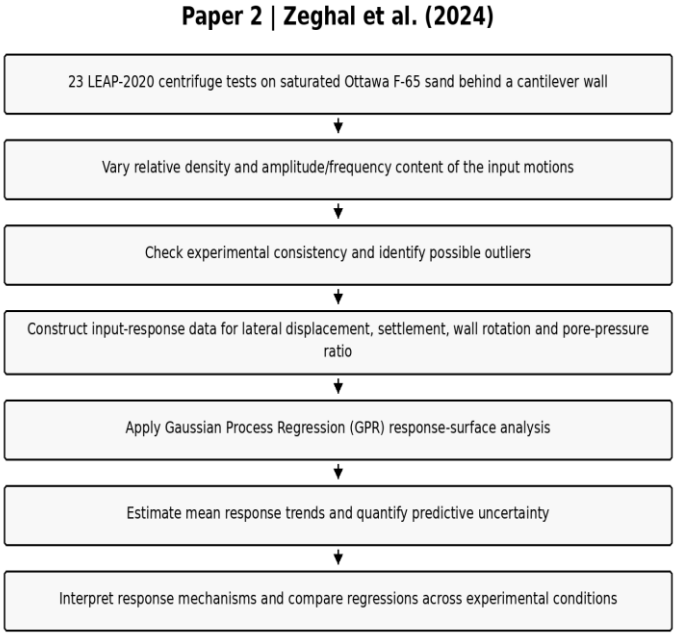


Figure 2.2 Methodology adopted in Paper 2 - Zeghal et al. (2024)

The important point for the present study is the decision to model engineering response quantities rather than simply treating thousands of time samples as independent rows. A similar first-stage model could relate sensor position, geometry and features of the box motion to peak or end-of-shaking pore pressure and acceleration, while the GP supplies an uncertainty range.

2.6 Paper 3 - The effects of base motion variability and soil heterogeneity on lateral spreading of mildly sloping ground

ElGhoraiby & Manzari (2020). The effects of base motion variability and soil heterogeneity on lateral spreading of mildly sloping ground.

The paper studies mildly sloping liquefiable ground and treats two major sources of variability: soil heterogeneity and the variability of achieved base motion. These sources are passed through coupled nonlinear finite-element calculations. Monte Carlo sampling is used to obtain a distribution of possible engineering responses and the resulting scatter is compared with centrifuge observations.

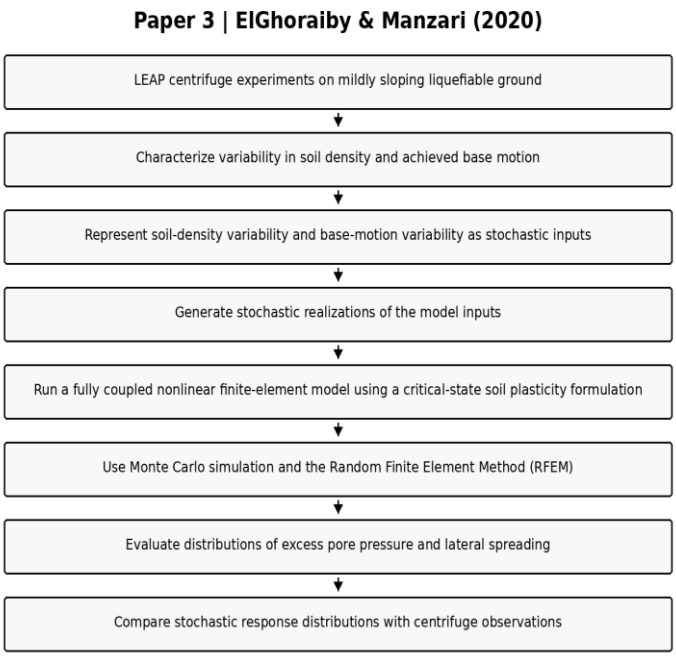


Figure 2.3 Methodology adopted in Paper 3 - ElGhoraiby & Manzari (2020)

The external accelerometer in the present experiment makes this approach particularly relevant. So, instead of viewing $a_{box}(t)$ as just another measured column, it can be treated as the excitation of the soil system. The paper supports the idea of analysing input variability separately from spatial soil variability before combining them in a response model.

2.7 Paper 4 - Stochastic analysis of the seismic performance of sheet-pile walls supporting heterogeneous liquefiable ground

ElGhoraiby & Manzari (2024). Stochastic analysis of the seismic performance of sheet-pile walls supporting heterogeneous liquefiable ground.

This study combines LEAP-2020 information, in-flight CPT data and achieved base motions. Relative-density variability is estimated from the CPT observations, while variability in the earthquake input is characterized through response spectra. The work therefore separates uncertainty in the soil from uncertainty in the applied motion.

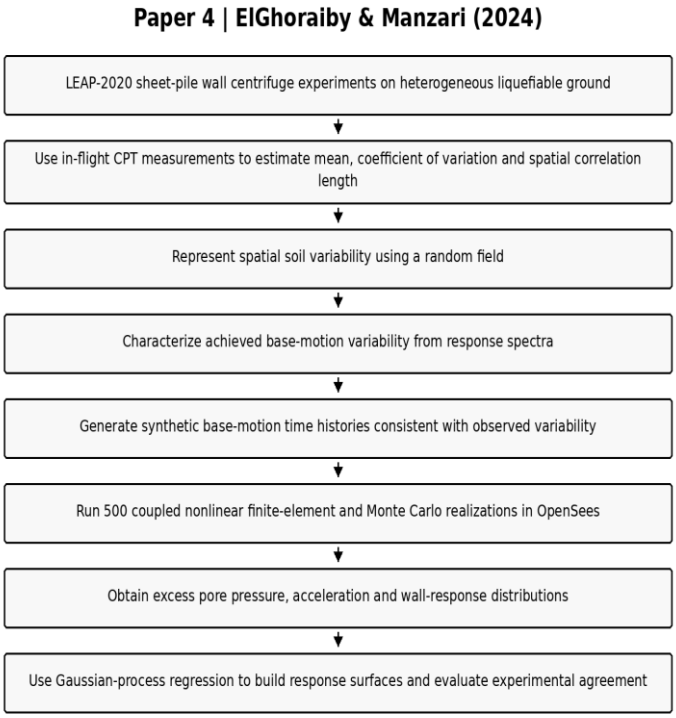


Figure 2.4 Methodology adopted in Paper 4 - ElGhoraiby & Manzari (2024)

Spatial random fields, synthetic base motions, nonlinear finite-element analysis and Monte Carlo simulation are combined. The authors also use a Gaussian-process response surface to summarize the resulting simulations. For the present problem, the useful lesson is the separation of input variability, spatial variability and response variability, with the external box accelerometer acting naturally as the input channel.

2.8 Paper 5 - Reliability assessment of the physical modeling of liquefaction-induced effects on shallow foundations considering nonuniformity in the centrifuge model

Kumar, Kasama & Takahashi (2020). Reliability assessment of the physical modeling of liquefaction-induced effects on shallow foundations considering nonuniformity in the centrifuge model.

The authors study a dynamic centrifuge model involving a liquefiable Toyoura sand deposit and a shallow foundation. An important observation is that a laboratory or centrifuge soil specimen

should not automatically be considered perfectly uniform. Spatial nonuniformity is represented statistically and then carried into numerical analyses.

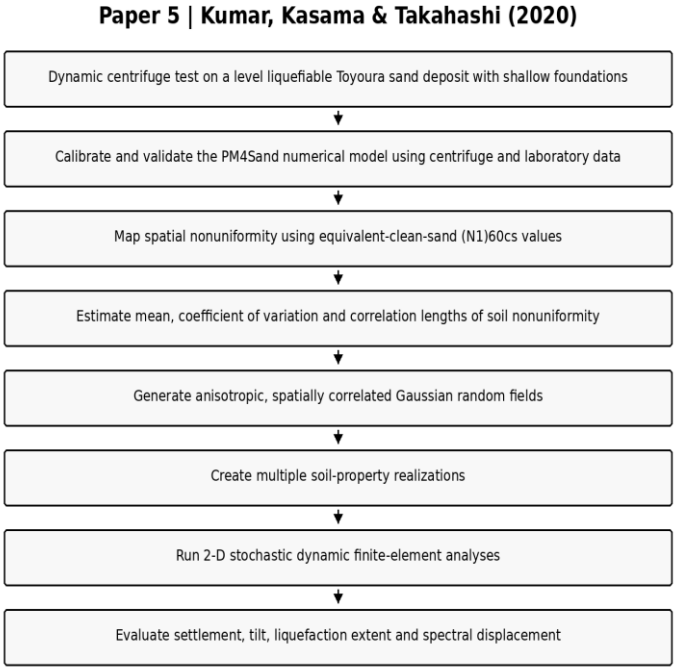


Figure 2.5 Methodology adopted in Paper 5 - Kumar, Kasama & Takahashi (2020)

The study uses an anisotropic, spatially correlated Gaussian random field and repeated finite-element analyses to evaluate distributions of settlement, tilt and liquefaction extent. The main lesson for the present test is that a spatial stochastic model needs measurable parameters such as mean, variance and correlation length. These should be estimated from the actual observations rather than selected arbitrarily.

2.9 Paper 6 - Effects of soil heterogeneity on the numerical simulation of liquefiable soil deposits using a multi-surface plasticity model

Mercado et al. (2026). Effects of soil heterogeneity on the numerical simulation of liquefiable soil deposits using a multi-surface plasticity model.

This recent study uses LEAP-2020 centrifuge information to investigate how spatial heterogeneity affects numerical predictions for liquefiable sand. A multi-surface plasticity

formulation is checked against the centrifuge response, after which relative-density variability is represented through a spatially correlated Gaussian random field.

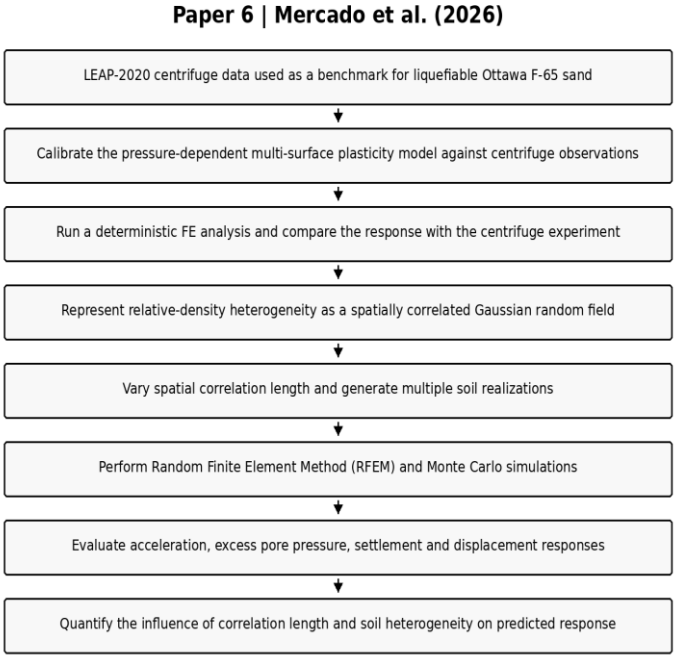


Figure 2.6 Methodology adopted in Paper 6 - Mercado et al. (2026)

Random finite element analysis and Monte Carlo simulation are used to generate response distributions for acceleration, excess pore pressure, settlement and displacement. The deliberate variation of correlation length is particularly relevant to the present work because the horizontal and inclined soil configurations may not share the same spatial dependence.

2.10 Paper 7 - Stochastic Modeling of SPT N-Value and Evaluation of Probability of Liquefaction at Guwahati City

Raghu Kanth & Dash (2008). Stochastic Modeling of SPT N-Value and Evaluation of Probability of Liquefaction at Guwahati City.

This paper is the clearest example in the selected set of generating a large amount of synthetic data from a limited database. SPT measurements from 62 boreholes are represented as a deterministic depth trend plus a stochastic residual. The fitted trend is $N_{av}(z) = 3.8208 +$

$1.1115z$, after which the residual is standardized and its spatial correlation is estimated from the autocorrelation function.

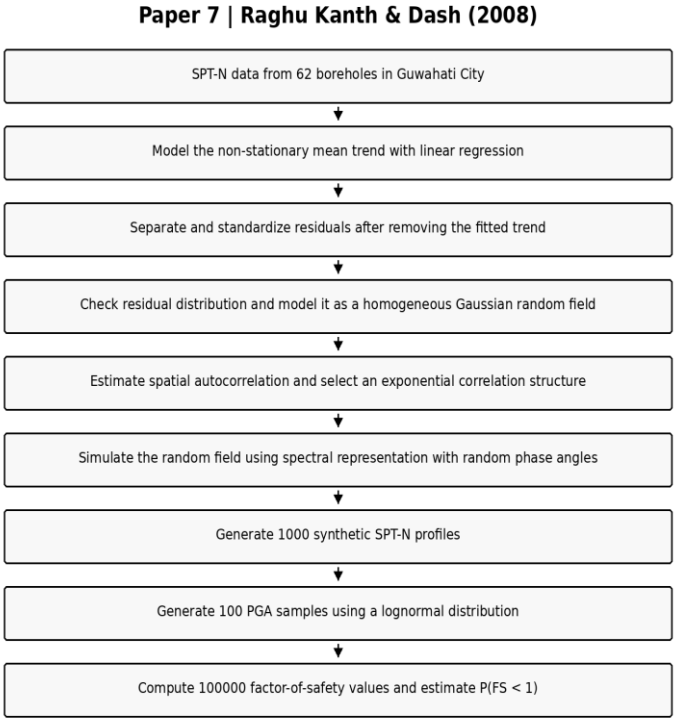


Figure 2.7 Methodology adopted in Paper 7 - Raghu Kanth & Dash (2008)

An exponential random field is selected and simulated using a spectral representation with random phases. The paper reports an estimated correlation distance of about 3.44 for the selected exponential form. Repeated simulations produce about 1000 synthetic SPT profiles. These profiles are combined with 100 lognormal PGA samples, leading to 100,000 simulated factor-of-safety values at each depth. Liquefaction probability is finally calculated from P(FS < 1). The main lesson for the present experiment is the sequence: remove trend, model residual correlation, generate realizations, restore the trend, and propagate the ensemble into an engineering calculation.

2.11 Comparative synthesis

The seven studies do not use stochastic modelling in exactly the same way. GP-based studies place the uncertainty in the input-response relationship and return a predicted response with uncertainty. Random-field studies describe spatial heterogeneity and then propagate it through a

numerical model. Stochastic base-motion studies add variability to the excitation itself. Monte Carlo appears mainly as a sampling layer that turns the defined probability model into an ensemble of possible cases.

Paper	Dataset / problem	Main stochastic method	Stochastic quantity	Main output
Goswami et al. [1]	LEAP-2017 sloping sand	GP-based kriging	Input and response differences	Sensitivity and uncertainty
Zeghal et al. [2]	23 LEAP-2020 centrifuges	GPR	Factors and response	Response trends and uncertainty
ElGhoraiby & Manzari [3]	Mildly sloping liquefiable ground	RFEM + MC	Soil and base motion	EPWP and spreading
ElGhoraiby & Manzari [4]	LEAP-2020 wall problem	RFEM + MC + GP	Soil and base motion	EPWP and acceleration
Kumar et al. [5]	Toyoura-sand centrifuge	GRF + FE	Spatial nonuniformity	Settlement, tilt, liquefaction extent
Mercado et al. [6]	LEAP-2020 liquefiable sand	GRF + RFEM + MC	Relative-density heterogeneity	Acceleration and EPWP
Raghu Kanth & Dash [7]	62 Guwahati boreholes	Random field + spectral simulation + MC	SPT-N residual and PGA	Synthetic profiles and P(FS<1)

For the present study, the most transferable ideas are not individual numerical values but the modelling sequence. The sensor data should first be cleaned and synchronized. Trends and distributions should then be examined. Spatial and temporal correlation should be estimated. Only after this should a stochastic model be fitted and used to generate conditional realizations. The generated ensemble must then be compared with the measured response statistics.

2.12 Research gap and relevance to the present study

The reviewed studies demonstrate strong precedents for stochastic modelling of soil heterogeneity, earthquake input motion and engineering response. However, the selected papers do not directly present a data-driven framework for generating spatially and temporally correlated sensor observations from a small set of synchronized pore-pressure and acceleration histories.

The present test has an additional feature in the external box accelerometer, which creates a clear input-output structure: $a_{box}(t)$ is the excitation, while a(x,y,z,t) and u(x,y,z,t) are responses. The second, inclined configuration also creates a reason to test directional or anisotropic correlation.

The next stage can therefore investigate whether a spatiotemporal GP which is possibly conditioned on the box motion and extended to joint pore-pressure and acceleration response, can

reproduce the measured statistics and generate plausible conditional realizations. Random-field and Monte Carlo methods can be used alongside this model when dense spatial fields or uncertainty propagation are required.

CHAPTER 3

PRESENT EXPERIMENT

3.1 Present centrifuge experiment

The experiment motivating the study is a centrifuge test on sand under earthquake or cyclic shaking. A rectangular soil box contains four levels of sensors. Each level has six sensor positions involving pore-pressure transducers and accelerometers. An additional accelerometer outside the box records the shaking of the container. The same overall test is repeated for an inclined arrangement of the levels after the horizontal case.

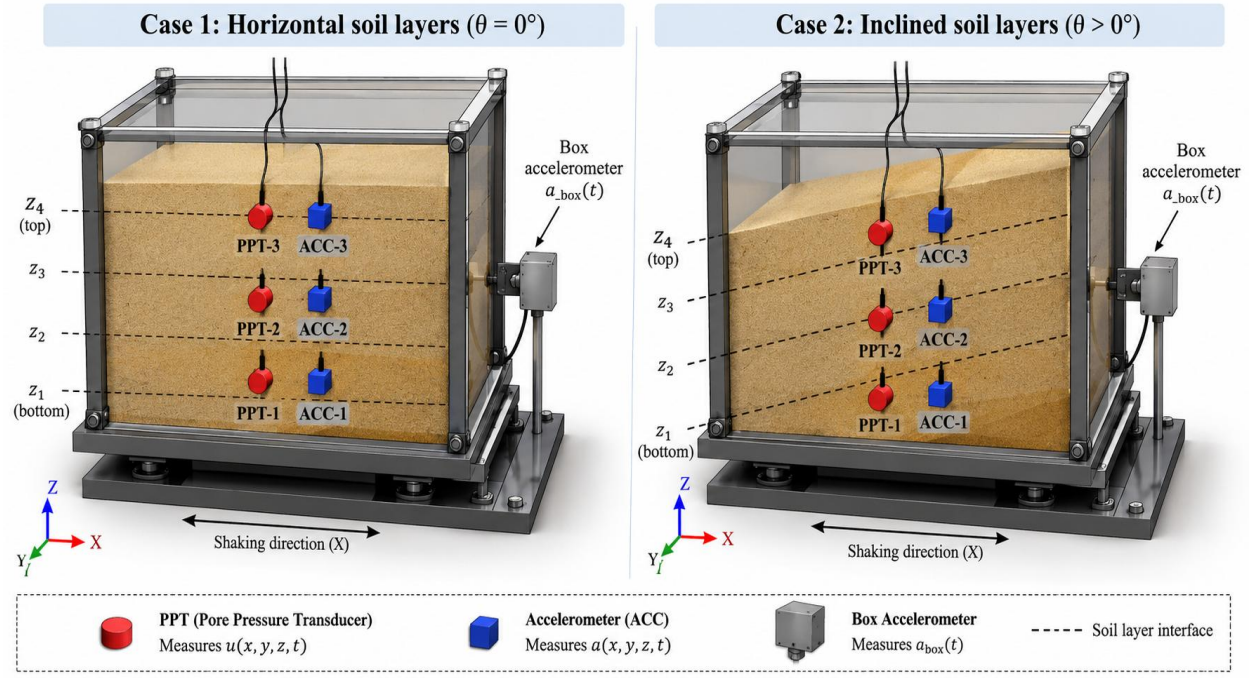


Figure 3.1 Schematic representation of the present centrifuge experiment and sensor arrangement

3.2 Sensor arrangement and measured quantities

Quantity	Notation	Role
Spatial position	x, y, z	Locates the sensor in the soil box
Time	t	Defines the measurement instant
Pore-water pressure	u(x,y,z,t)	Liquefaction-related response
Internal acceleration	a(x,y,z,t)	Dynamic soil response
Box acceleration	a_box(t)	Input / excitation record
Configuration	Case or θ	Horizontal or inclined geometry

The complete observation can therefore be represented conceptually as $D = \{x_i, y_i, z_i, t_j, u_{ij}, a_{ij}, a_{box}(tt_j), Case\}$.

3.3 Spatial and temporal structure

Measurements from nearby sensors are likely to be related because they sample a common soil mass. Measurements at similar depths can also share the same loading history and stress environment. In time, consecutive observations are strongly related. Therefore, a large row count does not mean a large number of independent observations.

This distinction is important for synthetic-data generation. Drawing independent values from a normal distribution can match a histogram while completely missing the correlation seen between neighbouring sensors or adjacent time samples. The stochastic model has to preserve the dependence structure.

3.4 Horizontal and inclined configurations

For the horizontal case, the natural coordinates are x, y and z. For the inclined case, the same coordinates can be retained, but an additional local coordinate system can be useful: s along the layer and n normal to it. If the response is more strongly correlated along the layer, the correlation length l_s may be different from l_n. The two test cases should first be analysed separately before a single combined model is considered.

3.5 Internship workflow

The internship had two connected components. The primary component was literature review. Each stochastic paper was read with the same questions in mind: what was uncertain, how was it quantified, how was a stochastic realization generated, and how was the result checked? Flowcharts were then prepared for each paper. The supporting component involved learning ANN, CNN, RNN, LSTM, BiLSTM and GRU concepts and practicing basic model training on standard datasets. This helped build the background needed to understand how a future stochastic database might be used by a data-driven model.

CHAPTER 4

MODELLING CONCEPTS AND PROPOSED FRAMEWORK

4.1 Why stochastic modelling is suitable

In this context, stochastic modelling means representing uncertainty together with structure. The structure can be a probability distribution, a covariance function, a correlation length, or a time-dependent random process. The objective is not to manufacture new experimental evidence but to draw plausible realizations from a model fitted to the information in the original experiment.

4.2 Gaussian Process regression

A Gaussian Process places a probability distribution over functions. It can be written as $f(r) \sim GP(m(r), K(r, r'))$, where m is the mean and K is the covariance kernel. A squared-exponential kernel is $K(r, r') = \sigma_f^2 \, exp(-(r - r')^2/(2l^2))$. The length scale l controls how quickly the correlation decreases with separation.

As a simple numerical example, consider three pore-pressure measurements at one depth: 42 kPa at x = 0.10 m, 46 kPa at x = 0.20 m and 51 kPa at x = 0.30 m. If $\sigma_f^2 = 100$ and l = 0.10 m, the covariance between 0.10 m and 0.20 m is 60.65, while the covariance between 0.10 m and 0.30 m is 13.53. At x = 0.25 m, the covariances with the three observations are about 32.47, 88.25 and 88.25, so the nearby observations receive more weight.

With $y = [42, 46, 51]^{\mathrm{T}}$, the predictive mean is $\mu_0 = k_0^T K^{-1} y$ and the predictive variance is $\sigma_0^2 = k(x_0, x_0) - k_0^T K^{-1} k_0$. The GP therefore gives both a central prediction and uncertainty. This is useful for a sensor location where no measurement is available, but the covariance structure must be estimated from the real experiment.

4.3 Spatiotemporal Gaussian Process

For the centrifuge data, the input to the GP can be $r = (x, y, z, t)$. A simple separable kernel is $K(r, r') = K_x \, K_y \, K_z \, K_t$. For example, $K_x = exp(-(x - x')^2/(2{l_x}^2))$ and $K_t = exp(-(t - t')^2/(2{l_t}^2))$. A pair of observations that are close in both space and time then receives a larger covariance than a pair that is far apart.

Suppose two observations are 0.05 m apart in x, 0.05 m apart in z and 0.01 s apart in time, with $l_x = l_z = 0.10\, m$ and $l_t = 0.02\, s$. Each corresponding kernel term is about 0.8825 and

the combined value is about 0.687. This is a simple demonstration of how space and time can be handled together.

The direct candidate models are $u(x,y,z,t) \sim GP(m_u, K_u)$ and $a(x,y,z,t) \sim GP(m_a, K_a)$. The empirical ACF and spatial covariance should be examined before choosing the kernel. The inclined case may require an anisotropic form because correlation can depend on direction.

4.4 Multi-output and input-conditioned GP

Pore pressure and acceleration are responses of the same soil system. A multi-output GP can use a block covariance matrix: $\begin{matrix} K_{uu} & K_{ua} \\ K_{au} & K_{aa} \end{matrix}$. This allows the model to use the statistical dependence between the two responses. The external accelerometer adds another layer: $a_{box}(t)$ can be used as an input, giving a conceptual model $u(x,y,z,t) = f[a_{box}(0{:}\,t), x, y, z, geometry] + \varepsilon$. The use of the motion history is important because soil response can depend on the loading that has already occurred, not only on instantaneous acceleration.

4.5 Gaussian Random Field and Sequential Gaussian Simulation

A Gaussian Random Field is useful when the objective is a spatial field rather than one missing point. The dependence can be described using a variogram: $\gamma(h) = \frac{1}{2N(h)} \Sigma_{i=1}^{N(h)} [z(xx_i) - z(xx_i + h)]^2$, or by a covariance function. Sequential Gaussian Simulation then uses the fitted spatial model to generate values at unmeasured locations while conditioning on the observed points. Repeating the process produces different spatial realizations. For an inclined layer, the field can be made anisotropic so that the correlation length along the layer differs from the correlation length normal to the layer.

4.6 Monte Carlo simulation

Monte Carlo is best treated as the sampling mechanism after a probability model has been defined. If X is stochastic, repeated samples $X_1, X_2, \ldots, X_N$ can be used to estimate averages or probabilities. The Raghu Kanth and Dash study is a useful example: 1000 synthetic SPT profiles were combined with 100 PGA samples to produce 100000 factor-of-safety values. In the present problem, Monte Carlo could similarly be used to sample multiple conditional GP or random-field realizations, but it should not be confused with the underlying correlation model.

4.7 Deep learning models studied

ANN was studied as a general nonlinear mapping for tabular geotechnical data. CNN was studied for its ability to detect local spatial patterns when a soil profile or response field is arranged as a grid. RNN, LSTM, BiLSTM and GRU were studied for ordered data. The main practical lesson was that these models require careful preprocessing and a genuinely independent validation set.

The sequence perspective may be useful later for sensor time histories, while CNNs may become relevant if the stochastic model produces a dense spatial field. These applications should come after the stochastic model has been validated, because increasing the number of synthetic rows does not automatically increase the amount of physical information.

4.8 Proposed combined framework

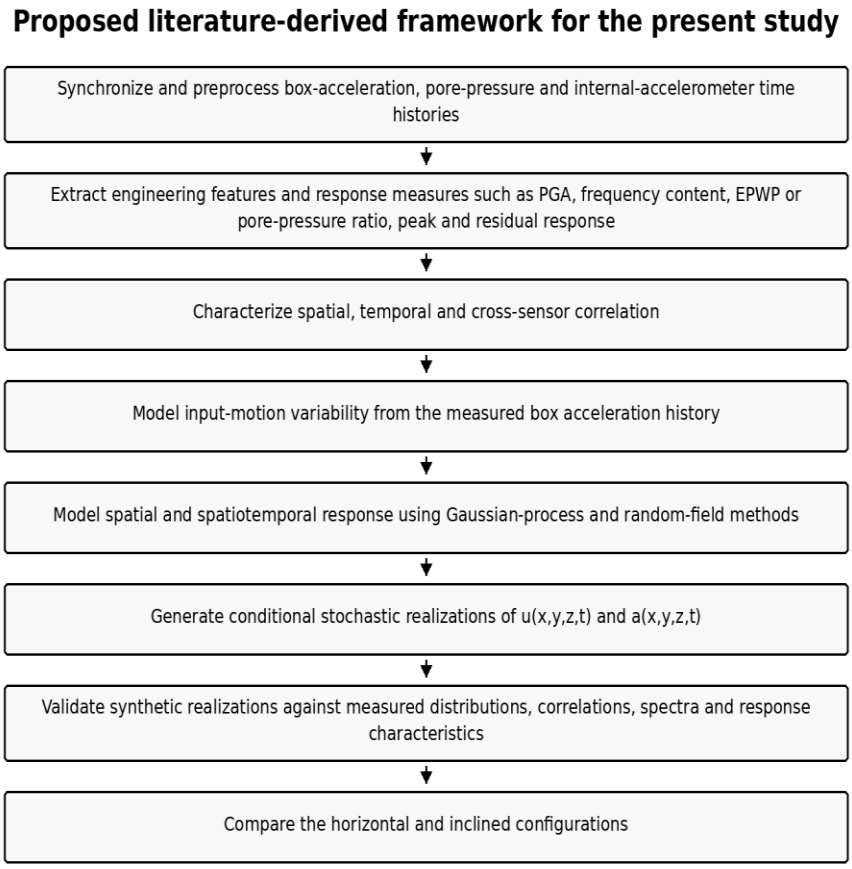


Figure 4.1 Proposed literature-derived stochastic framework for the present study

The proposed literature-derived framework starts with synchronized pore-pressure, internal acceleration and box-acceleration records. Engineering features and full time histories can be treated separately. Spatial and temporal dependence is then characterized, followed by fitting of a spatiotemporal GP or a random-field model. The fitted model is used to produce conditional stochastic realizations. These realizations are compared with the original experiment before they are considered suitable for any downstream machine learning application.

CHAPTER 5

APPLICATION AND DISCUSSION

5.1 Lessons from stochastic literature

The clearest lesson is that synthetic-data generation should start with statistical characterization. The Guwahati study first separated the depth trend from the residual variability. The centrifuge studies similarly distinguish soil heterogeneity from base-motion variability. For the present experiment, the first step should therefore be to identify trends, noise levels and correlation in each measured quantity.

The second lesson is that correlation length matters. Two datasets can have the same mean and variance while behaving differently if one is strongly correlated in space and the other is nearly independent. This is especially relevant for the inclined configuration, where directional dependence is physically plausible.

The third lesson is validation. A successful stochastic reconstruction should not be judged only by whether the synthetic histogram looks similar to the measured histogram. Temporal ACF, spatial correlation, cross-correlation between pore pressure and acceleration, peak response and frequency content should also be checked.

5.2 Lessons from deep learning practice

The deep learning exercises provided a complementary perspective. ANN can handle tabular predictors, CNN can exploit local spatial structure, and sequence models can retain information across ordered observations. The practical work also made the importance of normalization, validation and overfitting clear. For the present problem, a random split of individual time samples could leak information between training and testing because neighbouring samples belong to the same physical event. Future validation should therefore be organized by experiment or loading events whenever possible.

5.3 Mapping the literature to the present dataset

Finding from literature	Implication for present project
Input motion varies between nominally similar tests	Use a_box(t) explicitly and characterize its variability
Soil and response fields show spatial dependence	Estimate spatial correlation before generating values
GP can learn nonlinear relationships from limited experimental data	Use GPR for response surfaces and uncertainty
Random fields preserve spatial dependence	Use GRF or SGS when dense spatial realizations are required
Monte Carlo creates an ensemble after a probability model is defined	Use MC as a realization layer, not as the main model
Inclined ground can motivate directional correlation	Test anisotropic correlation in the inclined case
Validation must compare stochastic output with experiments	Use statistical, temporal, spatial and spectral checks

5.4 Validation and precautions

- Compare experimental and synthetic means and variances for pore pressure and acceleration.
- Compare temporal autocorrelation at several time lags and spatial correlation over sensor separation distances.
- Check cross-correlation between acceleration and pore pressure, including possible lag.
- Compare peak response, RMS response and relevant frequency or response-spectrum measures.
- For liquefaction interpretation, compare peak excess pore pressure and pore-pressure ratio together with the time at which the buildup occurs.
- Keep in mind that a synthetic dataset is a model-based ensemble, not a substitute for additional physical experiments.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

The summer internship established a practical understanding of how stochastic modelling has been used in geotechnical earthquake engineering and how those ideas relate to centrifuge experiments on liquefiable sand. Seven papers were reviewed in detail, and a flowchart was prepared for each so that the route from observations to stochastic output could be understood clearly.

The literature shows three useful modelling routes. GP-based methods are appropriate for learning nonlinear response relationships and uncertainty. Gaussian random fields and related spatial simulation methods are appropriate for representing spatial variability. Monte Carlo is useful for generating an ensemble of outcomes after the probability model has been defined. The selected papers also show that earthquake input motion can itself be treated as a source of uncertainty.

For the present experiment, the measured box acceleration provides a natural input, while internal acceleration and pore-pressure histories provide the response. The problem is therefore naturally written as $u(x, y, z, t)$ and $a(x, y, z, t)$ conditioned on $a_{box}(t)$ and the test geometry. The inclined case may require anisotropic spatial modelling. The combined literature suggests that a spatiotemporal GP, supported by random-field and Monte Carlo methods is a sound direction for the next stage.

6.2 Limitations

- The internship focused on literature review and theoretical methodology; the final stochastic model was not calibrated on the actual centrifuge sensor files in this report.
- No numerical values for the actual spatial or temporal correlation lengths of the test are reported because those parameters must be estimated from the experimental record.
- The deep learning part was introductory and was not trained on the centrifuge response dataset.

- The seven papers address related but different problems, so their assumptions need to be checked before direct transfer to the present experiment.

6.3 Future work

1. Obtain and synchronize the complete sensor dataset with exact coordinates and sampling information.
2. Estimate distributions, trends, spatial correlation and temporal ACF separately for the horizontal and inclined cases.
3. Fit a first-stage spatiotemporal GP for pore pressure and acceleration using the box motion as an input or through suitable motion descriptors.
4. Investigate anisotropic random fields for the inclined geometry and compare them with isotropic alternatives.
5. Generate conditional stochastic realizations and validate them against experimental statistics, spectra and engineering response measures.
6. Once the stochastic generator is validated, investigate whether the realizations can support a ML/DL model.

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